

Parametric Value at Risk(VaR) BackTest

STAT 649 Final Project

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1 Introduction

Value at Risk(VaR)is one of the most common risk measures used in finance. The correct estimation of VaR is essential for any financial institution, in order to arrive at the accurate capital requirements and to meet the adverse movements of the market.[1] Value-at-Risk is a measure of the maximum potential change in value of a portfolio of financial instruments with a given probability over a pre-set horizon.[2]

A derivative trading organization would need VaR for the following reason;

- Accurate capital requirement estimation. This is for compliance with regulatory authorities.
- Risk Management.
- Optimal portfolio construction.
- Scenario Analysis.
- Competitive Advantage
- Adaptation to market changes.

1.1 The Problem

Compare Parametric Value-at-Risk (VaR) using Simple Moving Average, Exponentially Weighted Moving Average (EWMA), and Generalized Auto-Regressive Conditional Heteroskedasticity (GARCH) to estimate the volatility. You can assume a managed future fund

with \$10M asset on Jan 1, 2000 and always longing or shorting 2nd nearby of WTI and Henry Hub with equal weights.

2 Literature Review

The literature on parametric Value at Risk (VaR) analysis is enriched by the contributions of several seminal papers, each shedding light on distinct aspects of this risk measurement methodology. In the work titled "Parametric Value at Risk Analysis: Evidence from Stock Indices" by Samir Mabrouk and Samir Saadi, the authors delve into the application of parametric VaR to stock indices, providing empirical evidence on its efficacy in capturing and quantifying market risk. Complementing this, Deepak Jadhav and T.V. Ramanatha, in their paper on "Parametric and Non-parametric Estimation of Value at Risk," contribute to the literature by comparing parametric and non-parametric approaches to VaR estimation. Their analysis offers valuable insights into the relative strengths and weaknesses of these methodologies, adding nuance to the ongoing debate on the most suitable techniques for risk measurement. Furthermore, J. Carlos Escanciano and Jose Olmo explore the critical dimension of backtesting parametric VaR in their paper "Backtesting Parametric VaR with Estimation Risk," examining the robustness of parametric VaR models to estimation risk. These contributions collectively form a comprehensive literature foundation, advancing our understanding of parametric VaR in different financial contexts and highlighting its implications for risk management practices.

3 Analysis

There are two predominant approaches to calculating VaR, parametric and non-parametric, have emerged in the financial literature to address the complexities of VaR estimation. The parametric method relies on specific distributional assumptions, such as normal or student's t-distributions, to model the underlying risk factors. On the other hand, the non-parametric approach dispenses with explicit distributional assumptions, allowing for greater flexibility in capturing the inherent complexities and irregularities of financial markets.

Non-parametric methods, including historical simulation and kernel density estimation, leverage observed data directly, making them robust in handling various types of market conditions. In this project we will focus on parametric methods to calculate VaR.[3]

3.1 Method

The method we will focus on is from [2] and is outlined as follows,

- **Step 1** Portfolio marked to market, $\mathbf{V}_0$
- **Step 2** 1 day forecast of return $\mu_{1|0}$. 1 day horizon we can assume mean return is zero.
- **Step 3** Estimate $\sigma_{1|0}$, the standard deviation of the portfolio. Parametric VaR assumes normal distribution of returns as $\hat{r} = -1.65 * \sigma_{1|0} + \mu_{1|0}$, thus $\mathbf{V}_1 = \mathbf{V}_0 * (1 - \hat{r})$.
- **Step 4** Value at Risk is given by $\mathbf{V}_0 - \mathbf{V}_1$.

We are going to use 3 method to estimate volatility in step 3 as follows,

3.1.1 Simple Moving Average(SMA)

SMA calculates the average volatility of a security or asset over a set number of periods by summing up the returns and dividing them by the chosen time frame. All the returns are weighted equally in this method. The formula is as follows

$$\sigma^2 = \frac{1}{n-1} \sum_{t=1}^n (R_t - \mathbf{E}(R_t))^2 \quad (1)$$

thus volatility estimate is

$$\sigma_{SMA} = \sqrt{\sigma^2}$$

3.1.2 Exponential Weighted Moving Average(EWMA)

Unlike SMA, EWMA assigns exponentially decreasing weights to past observations, with more recent data receiving higher significance. This characteristic allows the EWMA to be more responsive to recent price changes, making it particularly useful for capturing evolving trends and adapting to market volatility. The choice of the smoothing parameter, often denoted as the lambda (λ) parameter, influences the rate at which older observations contribute to the average. In practise $\lambda = 0.94$ for daily data and $\lambda = 0.97$ for monthly data. The formula is as follows,

$$\sigma^2 = \sum_{t=1}^n (1-\lambda)\lambda^{t-1} R_t^2 \quad (2)$$

thus volatility estimate is

$$\sigma_{EWMA} = \sqrt{\sigma^2}$$

3.1.3 Generalized Auto-Regressive Conditional Heteroskedasticity (GARCH)

Generalized Auto-Regressive Conditional Heteroskedasticity (GARCH) is a statistical model commonly employed to analyze and model volatility clustering in time series data. Developed as an extension of the ARCH (Auto-Regressive Conditional Heteroskedasticity) model, GARCH introduces a more comprehensive and flexible framework for capturing the dynamic nature of volatility. GARCH models allow for the estimation of conditional variances, where the volatility at any given time is dependent on past squared observations and past conditional variances. This modeling approach is particularly relevant in finance, where volatility tends to exhibit clustering behavior, with periods of high volatility often followed by similar periods. The formula is as follows,[3]

$$\sigma_t^2 = \omega + (1 - \lambda)\epsilon_t^2 + \lambda\sigma_{t-1}^2 \quad (3)$$

thus volatility estimate is

$$\sigma_{GARCH} = \sqrt{\sigma^2}$$

3.2 Assumptions

In this project we are to either long or short the 2nd nearby for West Texas Intermediate (WTI) and Henry Hub (NG). Our portfolio will have equal weights for WTI and NG that is in January we are to long march contracts for WTI and NG. At or on the last trading day for the January contract month we sell the March contract and buy April contract which is now the second nearby. Thus we need to re-balance the portfolio at each last day of trading. To simplify this we decided to use the time series data for WTI and NG that is use the contract months in succession.

3.3 Data

We used the time series data for WTI and NG from Yahoo finance with contract months in succession as shown below,[4]

	WTI.Adjusted	NG.Adjusted
2005-01-31	-0.0162159591	0.003926729
2005-02-01	0.0005491082	0.002607559
2005-02-02	0.0000000000	0.024707419
2005-02-03	0.0021969634	0.019035484
2005-02-04	-0.0027397511	-0.019925262
2005-02-07	0.0021978224	-0.011435791

Figure 1: Snapshot of DATA used

Our portfolio consists of equal weights of WTI and NG and the daily returns are as follows,

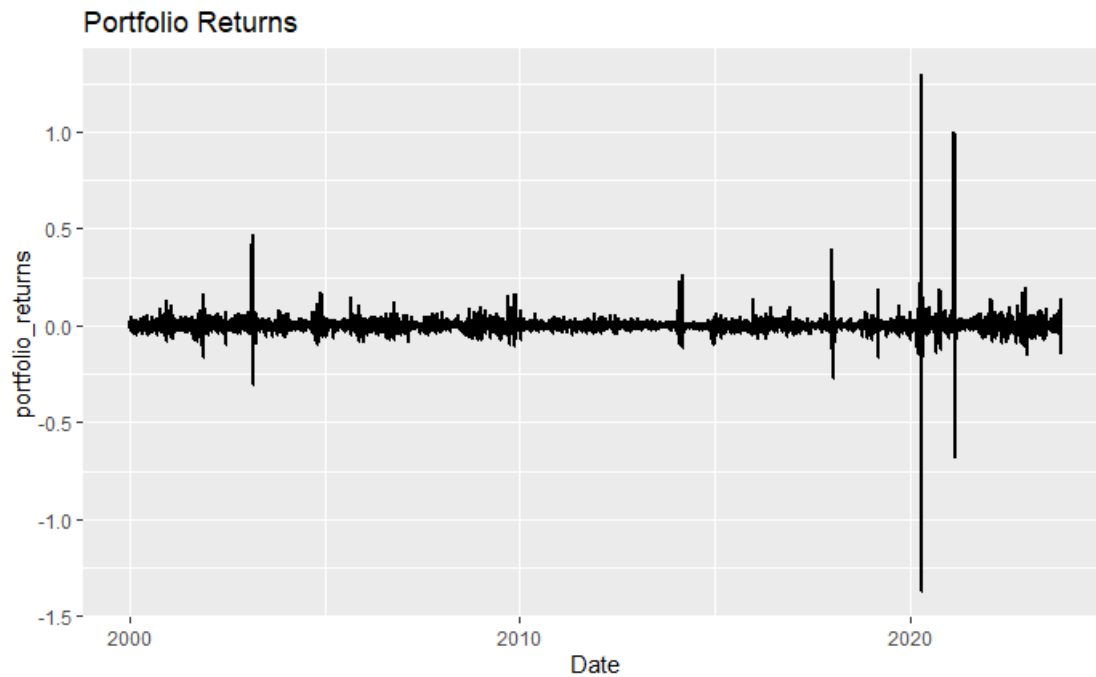


Figure 2: Portfolio Daily Returns

4 Results

Our comprehensive analysis focused on assessing the performance of three different volatility estimation methods in calculating VaR for individual assets and a diversified portfolio. The methods under scrutiny were SMA, EWMA, GARCH model. We will start by showing the volatility estimates from the different estimation methods

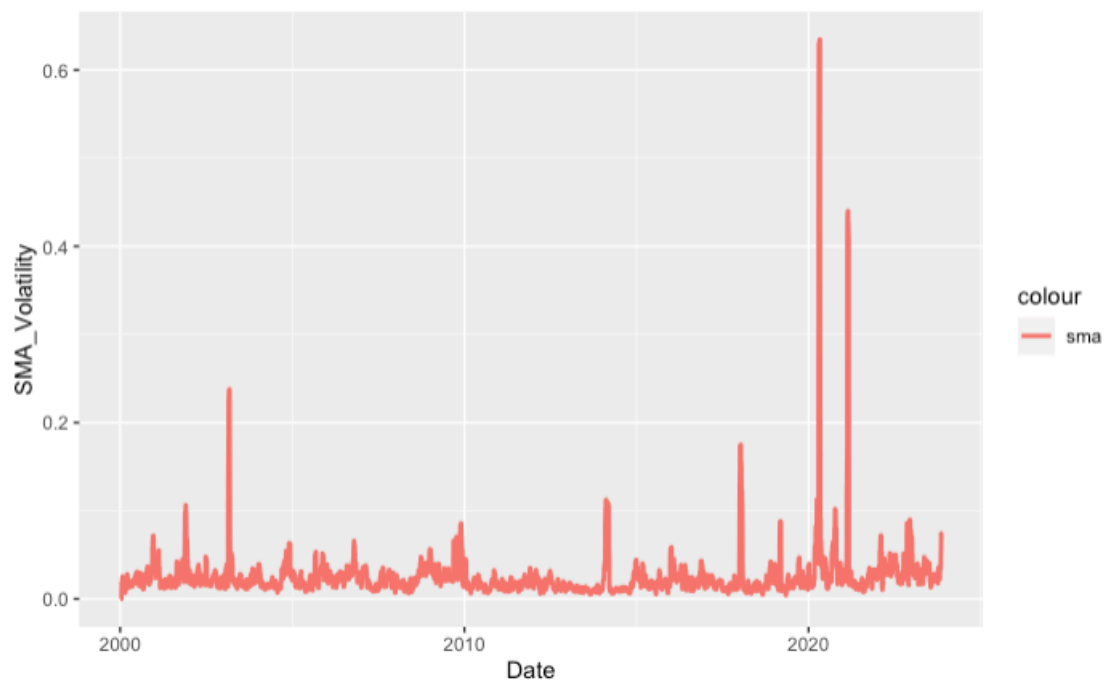


Figure 3: Volatility Estimate Using SMA

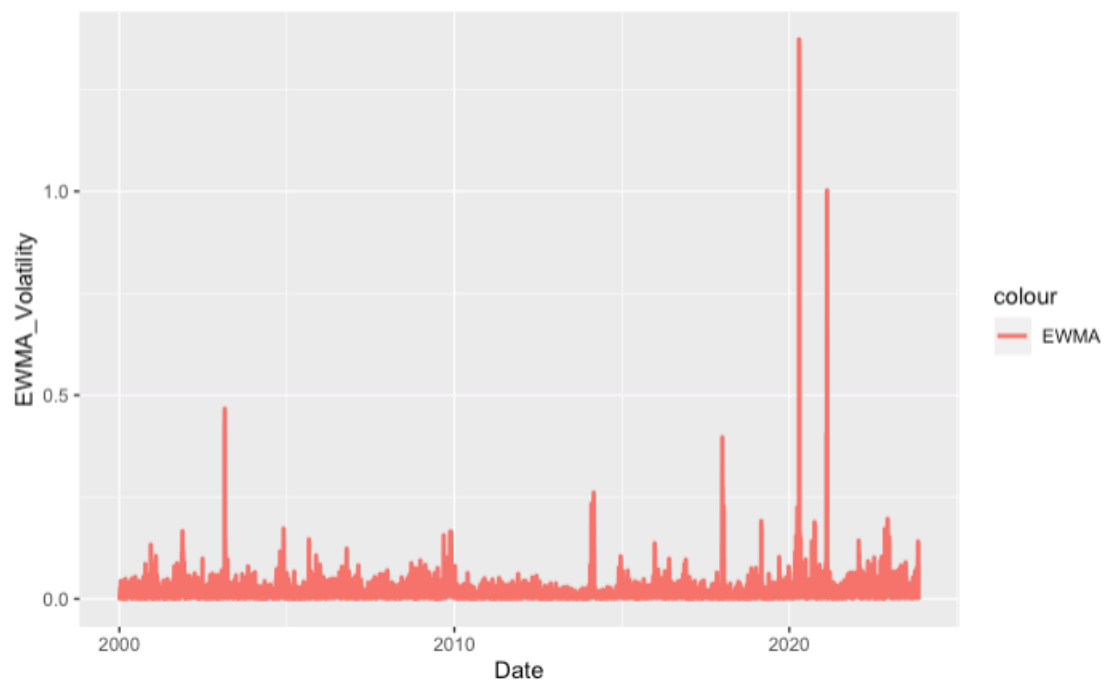


Figure 4: Volatility Estimate Using EWMA

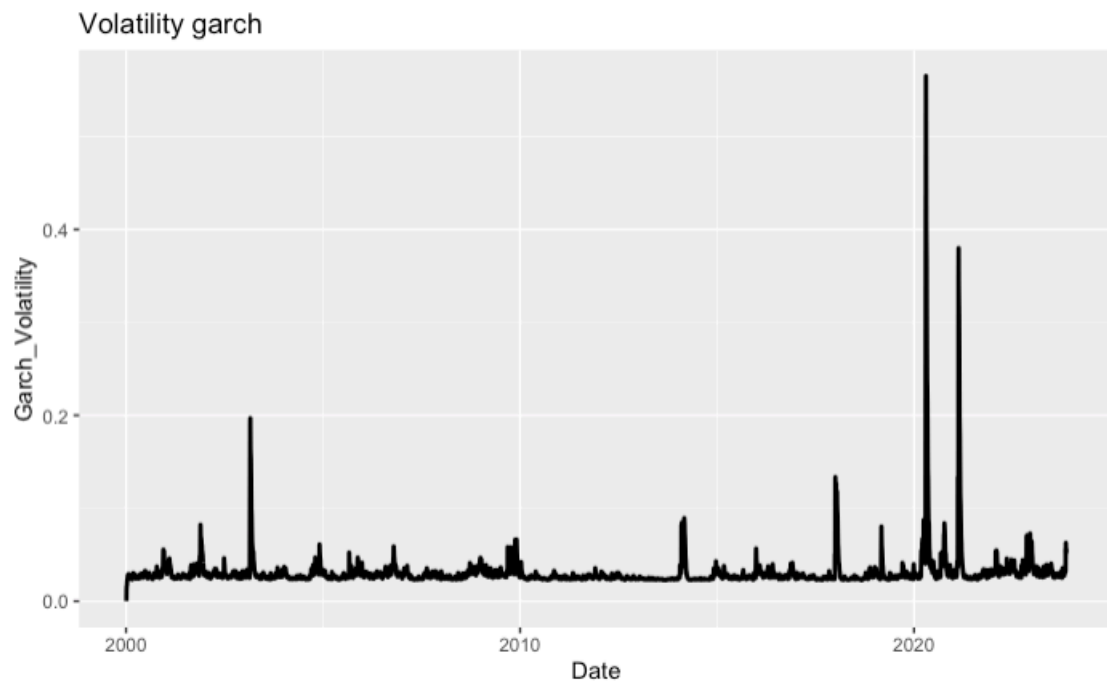


Figure 5: Volatility Estimate Using GARCH

The estimates of volatility's of WTI in Figures 3, 4,5 show a spike in 2020 when WTI opened at \$18 a barrel and fell to $-\$40$ the closed at $-\$37$. Volatility estimates decrease from SMA to GARCH.

Upon comparing the volatility estimates, SMA provided the highest values, potentially over-estimating risk due to its equal weighting of historical data. EWMA yielded higher estimates than GARCH but lower than SMA, striking a balance between responsiveness to recent data and historical trend consideration. GARCH estimates were the lowest, suggesting a more refined model that captures the intricacies of market behavior. We went and further calculated the VaR for each asset in the portfolio, and VaR was highest for WTI than NG because of higher volatility of WTI than NG.

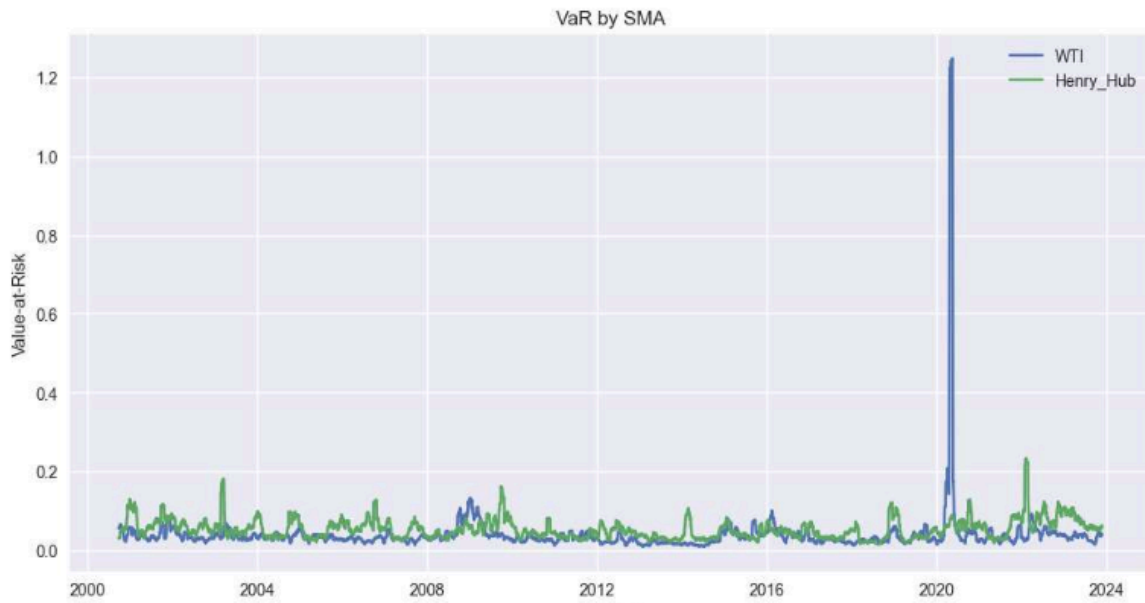


Figure 6: VaR for Individual Assets

The portfolio VaR is shown for the different methods is shown below

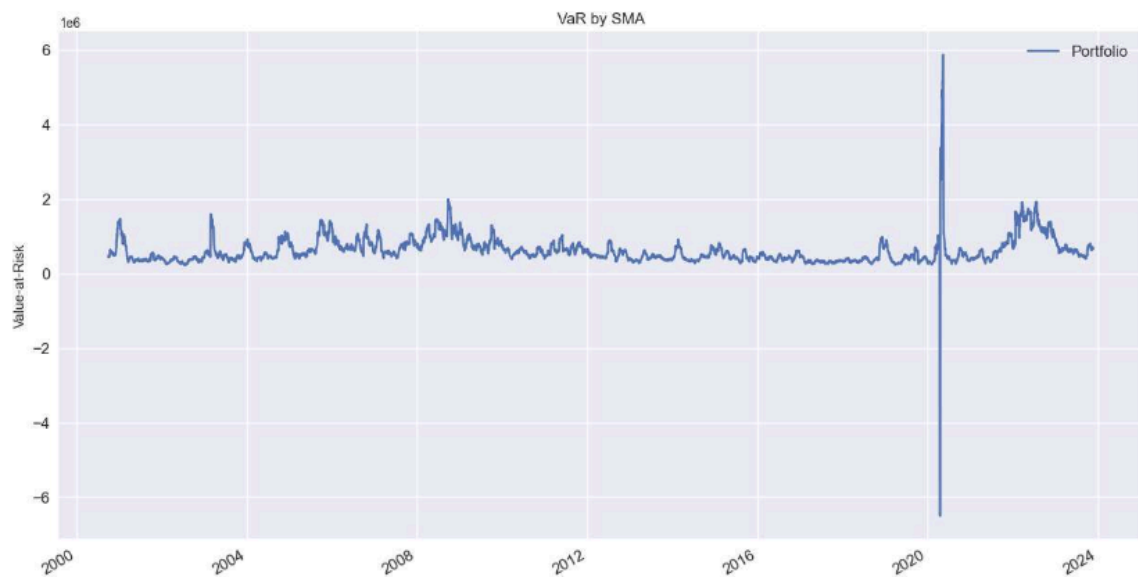


Figure 7: Portfolio VaR using SMA

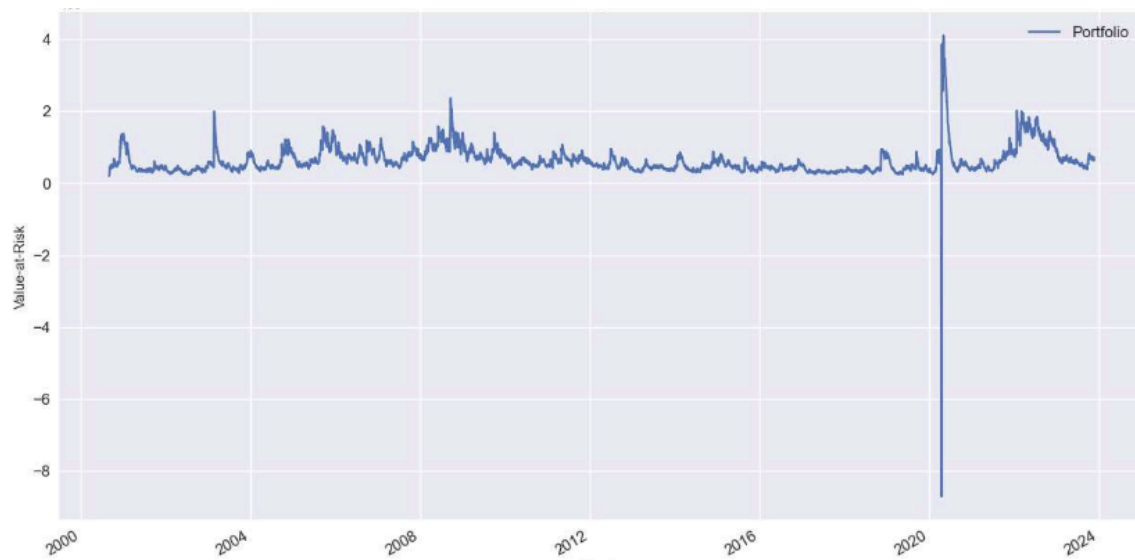


Figure 8: Portfolio VaR using EWMA

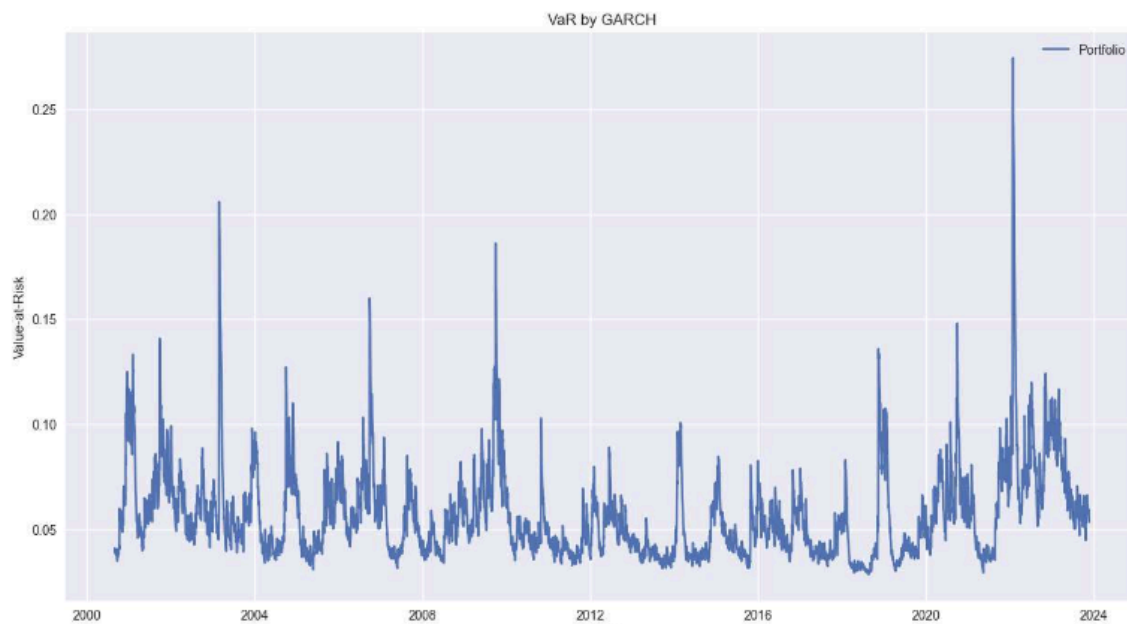


Figure 9: Portfolio VaR using GARCH

For the portfolio VaR, the SMA, EWMA, and GARCH models painted differing pictures of risk. SMA depicted a stable yet occasionally unresponsive risk profile, as evidenced

by the pronounced spike around 2020. In comparison, the EWMA model, which gives more significance to recent price movements, demonstrated heightened sensitivity to market volatility, resulting in a less dramatic spike during the same period.

We performed some test to compare the volatility estimates, we used the Root Mean Squared Error(RMSE) test to evaluate models and the result showed that the GARCH model had the lowest RMSE, indicating the highest level of accuracy in predicting market risk, followed by EWMA and SMA, respectively.

The back testing results were consistent with the RMSE findings. The GARCH model registered the least number of violations, followed by EWMA and SMA.[5] This implies that the GARCH model's predictions were the closest to the actual market movements, demonstrating its robustness in risk prediction. However the simplification of the data set caused our results to be less desirable than we expected. A managed fund of \$10M at the beginning of the period and is long our portfolio would have declined by about 10% to date as shown below

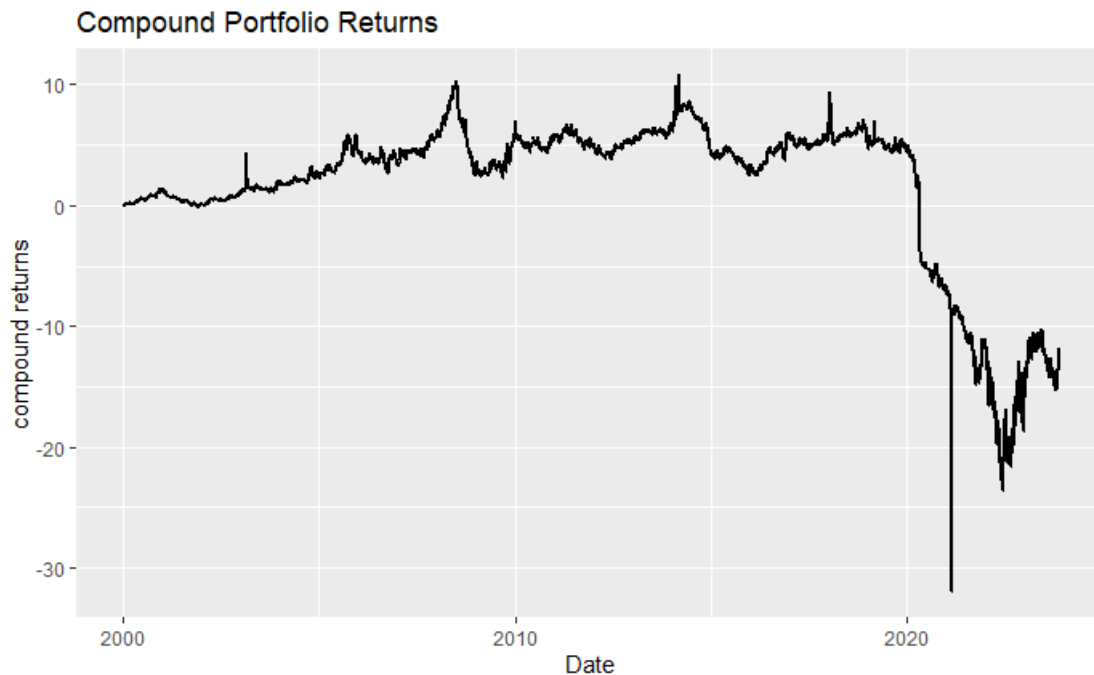


Figure 10: Long Portfolio Performance

5 Conclusion

The GARCH model outperformed SMA and EWMA in terms of predictive accuracy and back testing results. It demonstrated a superior ability to adapt to market volatility, which makes it a valuable tool for risk assessment in dynamic market conditions. While the SMA provides a more generalized and stable risk estimate, it may not be suitable for portfolios that require quick adaptation to market movements. EWMA serves as an intermediary, offering more sensitivity to recent events than SMA, but less than GARCH. Further research may explore using the exact 2nd nearby data of both WTI and NG. We could use FIGARCH or HYGARCH models discussed in [6] to have better results.

References

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